

```

1 def karatsuba(x, y):
2     if x < 10 or y < 10:
3         return x * y
4
5     n = max(len(str(x)), len(str(y)))
6     m = n // 2
7
8     high1 = x // (10 ** m)
9     low1 = x % (10 ** m)
10
11    high2 = y // (10 ** m)
12    low2 = y % (10 ** m)
13
14    z0 = karatsuba(low1, low2)
15    z1 = karatsuba(low1 + high1, low2 + high2)
16    z2 = karatsuba(high1, high2)
17
18    return z2 * (10 ** (2 * m)) + (z1 - z2 - z0) * (10 ** m) + z0
19
20
21 x = int(input())
22 y = int(input())
23
24 print("Product =", karatsuba(x, y))

```

input

1234
5678
Product = 7006652

...Program finished with exit code 0
Press ENTER to exit console.

```

6     if x < 10 or y < 10:
7         count += 1
8         return x * y
9
10    n = max(len(str(x)), len(str(y)))
11    m = n // 2
12
13    high1 = x // (10 ** m)
14    low1 = x % (10 ** m)
15
16    high2 = y // (10 ** m)
17    low2 = y % (10 ** m)
18
19    z0 = karatsuba(low1, low2)
20    z1 = karatsuba(low1 + high1, low2 + high2)
21    z2 = karatsuba(high1, high2)
22
23    return z2 * (10 ** (2 * m)) + \
24           (z1 - z2 - z0) * (10 ** m) + z0
25
26
27    x = int(input())
28    y = int(input())
29
30    result = karatsuba(x, y)
31
32    print("Product =", result)
33    print("Recursive Multiplications =", count)

```

input

```

1234
5678
Product = 7006652
Recursive Multiplications = 11

...Program finished with exit code 0
Press ENTER to exit console.

```

```

11     m = n // 2
12
13     high1 = x // (10 ** m)
14     low1 = x % (10 ** m)
15
16     high2 = y // (10 ** m)
17     low2 = y % (10 ** m)
18
19     z0 = karatsuba(low1, low2)
20     z1 = karatsuba(low1 + high1, low2 + high2)
21     z2 = karatsuba(high1, high2)
22
23     return z2 * (10 ** (2 * m)) + \
24           (z1 - z2 - z0) * (10 ** m) + z0
25
26
27 x = int(input())
28 y = int(input())
29
30 result = karatsuba(x, y)
31
32 n = max(len(str(x)), len(str(y)))
33
34 standard = n * n
35
36 print("Karatsuba Product =", result)
37 print("Karatsuba Multiplications =", count)
38 print("Standard Multiplications =", standard)

```

input

5678

```

Karatsuba Product = 7006652
Karatsuba Multiplications = 11
Standard Multiplications = 16

```

```

...Program finished with exit code 0
Press ENTER to exit console.

```

```

1 def karatsuba(x, y):
2     if x < 10 or y < 10:
3         return x * y
4
5     n = max(len(str(x)), len(str(y)))
6     m = n // 2
7
8     high1 = x // (10 ** m)
9     low1 = x % (10 ** m)
10
11    high2 = y // (10 ** m)
12    low2 = y % (10 ** m)
13
14    z0 = karatsuba(low1, low2)
15    z1 = karatsuba(low1 + high1, low2 + high2)
16    z2 = karatsuba(high1, high2)
17
18    return z2 * (10 ** (2 * m)) + \
19           (z1 - z2 - z0) * (10 ** m) + z0
20
21
22 x = int(input())
23 y = int(input())
24
25 print("Product =", karatsuba(x, y))

```



12345678

87654321

Product = 1082152022374638

...Program finished with exit code 0

Press ENTER to exit console.

```

11 high1 = x // (10 ** m)
12 low1 = x % (10 ** m)
13
14 high2 = y // (10 ** m)
15 low2 = y % (10 ** m)
16
17 z0 = karatsuba(low1, low2)
18 z1 = karatsuba(low1 + high1, low2 + high2)
19 z2 = karatsuba(high1, high2)
20
21 return z2 * (10 ** (2 * m)) + \
22        (z1 - z2 - z0) * (10 ** m) + z0
23
24
25 x = int(input())
26 y = int(input())
27
28 start = time.time()
29 result1 = x * y
30 standard_time = time.time() - start
31
32 start = time.time()
33 result2 = karatsuba(x, y)
34 karatsuba_time = time.time() - start
35
36 print("Product =", result1)
37 print("Standard Time =", standard_time)
38 print("Karatsuba Time =", karatsuba_time)

```

input

```

654321
Product = 80779853376
Standard Time = 2.86102294921875e-06
Karatsuba Time = 6.175041198730469e-05

```

```

...Program finished with exit code 0
Press ENTER to exit console.

```